TASK SPECIFICATION: P9-02 PHASE P9 — PRODUCTION DEPLOYMENT, CI/CD & OPERATIONS

Continuous Deployment (CD) Pipeline & Staging Delivery

Git Branch: `ci/P9-02-cd` · Target Pull Request: ci/P9-02-cd

Core Objective & Technical Scope:
Automated GitHub Actions deployment pipeline building verified production images and deploying to staging environment.

Priority Level	High	Complexity Rating	`M`
Execution Status	Backlog 📋	Target PR / Commit	`ci/P9-02-cd`
Upstream Dependencies	P9-01		
Architecture Blockers	None (Ready for execution)		

IMPLEMENTATION ROADMAP & ENGINEERING STEPS

1. Configure GitHub Actions CD workflow triggered on main branch tags.
2. Build and tag multi-arch production Docker images.
3. Deploy to staging target environment via SSH/Webhook.
4. Execute automated smoke verification tests post-deployment.

TARGET FILE INVENTORY & EXPECTED ARTIFACTS

- `.github/workflows/cd.yml`

ACCEPTANCE CRITERIA & QUALITY CHECKLIST

- Automated deployment completes without manual intervention.
- Failing smoke test triggers automatic deployment rollback.

VERIFICATION & TEST COVERAGE STRATEGY

- CD pipeline execution drill
- Post-deploy smoke test